

JEE MAIN REVISION MASTER: SETS AND RELATIONS

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 117 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Equivalence Relations (Reflexive, Symmetric, Transitive)	100 PYQs	Relations
Very High	Set Operations & Venn Diagram Principle	17 PYQs	Set Theory

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Equivalence Relations (Reflexive, Symmetric, Transitive)	$\text{Reflexive: } \forall a \in A, (a,a) \in R, \quad \text{Symmetric: } (a,b) \in R \implies (b,a) \in R, \quad \text{Transitive: } (a,b), (b,c) \in R \implies (a,c) \in R$
Set Operations & Venn Diagram Principle	$n(A \cup B) = n(A) + n(B) - n(A \cap B), \quad n(A \cup B \cup C) = \sum n(A) - \sum n(A \cap B) + n(A \cap B \cap C)$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Equivalence Relations (Reflexive, Symmetric, Transitive) (Very High Yield — 100 PYQs)

Core Principle: Formal definitions of binary relations on sets, properties for equivalence, and total number of reflexive relations $2^{n^2} - n$.

■ **Caution / JEE Pitfall:** A relation is vacuously transitive if there are no pairs (a,b) and (b,c) with distinct elements to contradict transitivity.

■ **Speed Trick:** Total number of relations on set of n elements is 2^{n^2} ; total number of symmetric relations is $2^{n(n+1)/2}$.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Consider the relation R on the set $-2,-1,0,1,2$ defined by $(a,b) \in R$ if and only if $1+ab>0$. Then, among the statements : I. The ... " [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "The number of elements in the relation $R=(x,y):4x^2+y^2<52,x,y \in Z$ is..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "The number of relations, defined on the set a,b,c,d , which are both reflexive and symmetric, is equal to:..." [Ans: N/A]

3.2 Set Operations & Venn Diagram Principle (Very High Yield — 17 PYQs)

Core Principle: Inclusion-exclusion principle for cardinality, De Morgan's set identities, and power set cardinality 2^n .

■ **Caution / JEE Pitfall:** If set A has n elements, the number of proper subsets is $2^n - 1$ (excluding the set itself).

■ **Speed Trick:** Number of elements belonging to EXACTLY two of three sets: $\sum n(A \cap B) - 3n(A \cap B \cap C)$.

Representative Source PYQs:

- **[2026 Evening Shift] Q#3 (MCQ):** "Let $A = \{x : |x^2 - 10| \leq 6 \text{ and } B = \{x : |x - 2| > 1\}$. Then..." **[Ans: B]**
- **[2025 Evening Shift] Q#3 (MCQ):** "Let $A = \{(\alpha, \beta) \in \mathbb{R} \times \mathbb{R} : |\alpha - 1| \leq 4 \text{ and } |\beta - 5| \leq 6\}$ and $B = \{(\alpha, \beta) \in \mathbb{R} \times \mathbb{R} : \alpha + \beta = 10\}$." **[Ans: N/A]**
- **[2025 Morning Shift] Q#3 (Numerical):** "For $n \geq 2$, let S_n denote the set of all subsets of $1, 2, \dots, n$ with no two consecutive numbers. For example $\{1, 3, 5\} \in S_6$." **[Ans: 13]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 22nd January Evening Shift (Q#1)

Let S be the set of the first 11 natural numbers. Then the number of elements in $A = \{B \subseteq S : n(B) \geq 2 \text{ and the product of all elements of B is even is } \underline{\hspace{2cm}}\}$.

Correct Answer: 1979

Solution Analysis: Let $S = \{1, 2, \dots, 11\}$. The product of all elements of B is even iff B contains at least one even number.
 Evens in S : 2, 4, 6, 8, 10. Odds in S : 1, 3, 5, 7, 9, 11. Number of subsets containing at least one even: $2^{11} - 2^6 = 2048 - 64 = 1984$ (we subtract the subsets made only of odds). Now enforce $n(B) \geq 2$: subtract the 5 singleton subsets 2, 4, 6, 8, 10. $1984 - 5 = 1979$

Benchmark Problem 2: JEE Main 2026 (Online) 21st January Evening Shift (Q#3)

Let $A = \{x : |x^2 - 10| \leq 6\}$ and $B = \{x : |x - 2| > 1\}$. Then

- (A) $A \cup B = (-\infty, 1] \cup (2, \infty)$
- (B) $B - A = (-\infty, -4) \cup (-2, 1) \cup (4, \infty)$ **[Correct]**
- (C) $A - B = [2, 3]$
- (D) $A \cap B = [-4, -2] \cup [3, 4]$

Correct Answer: B

Solution Analysis: For set A, we have the inequality: $|x^2 - 10| \leq 6$. This means that the expression inside the modulus lies between -6 and 6: $-6 \leq x^2 - 10 \leq 6$. By adding 10 to all parts of the inequality, we get: $4 \leq x^2 \leq 16$. Taking the square root gives us the range for $|x|$: $2 \leq |x| \leq 4$. This tells us that the value of x can be positive or negative, giving us the intervals: $A = [-4, -2] \cup [2, 4]$. Now, let's find the elements of set B from its inequality: $|x - 2| > 1$. This breaks down into two conditions: $x - 2 < -1$ or $x - 2 > 1$. Solving these, we get: $x < 1$ or $x > 3$. Writing this